AP PRECALCULUS

Unit 1 • Lesson 1.2 practice • Answer key

Lesson 1.2 • Rates of Change

5 multiple-choice questions and 1 free-response question • Answer key with explanations

Directions

Questions 1 to 5 are multiple choice: choose the best of the four answer choices. Question 6 is free response: show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

What is the average rate of change of $f(x) = 2x^2 - 3$ on $[1, 4]$?

(A) 10

(B) 15

(C) 30

(D) 8

Solution

Answer: (A). $f(4) = 29$, $f(1) = -1$, so $\dfrac{29 - (-1)}{3} = 10$.

Question 2

NO CALCULATOR

What is the slope of the secant line to $f(x) = x^3$ through the points where $x = -1$ and $x = 3$?

(A) $\dfrac{26}{4}$

(B) 7

(C) 9

(D) 28

Solution

Answer: (B). $\dfrac{27 - (-1)}{3 - (-1)} = \dfrac{28}{4} = 7$.

Question 3

NO CALCULATOR

$V(t)$ is the volume of water in a tank, in liters, t minutes after it starts filling. What are the units of the average rate of change of V?

(A) Minutes per liter

(B) Liters

(C) Liters per minute

(D) Liters times minutes

Solution

Answer: (C). Output units per input unit: liters per minute.

Question 4

NO CALCULATOR

Using the interval $[2,\ 2.001]$, which is the best estimate of the rate of change of $f(x) = x^2 + x$ at $x = 2$?

(A) 4
(B) 6
(C) 2.5
(D) 5

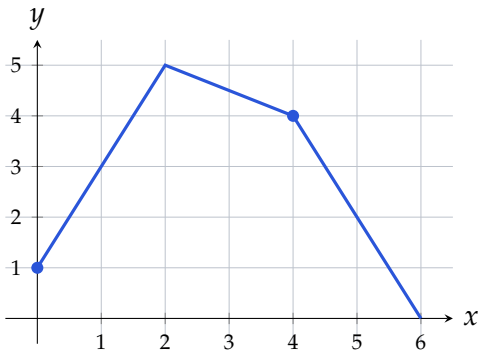
Solution

Answer: (D). $\dfrac{f(2.001) - f(2)}{0.001} = \dfrac{6.005001 - 6}{0.001} = 5.001 \approx 5.$

Question 5 (graph)

NO CALCULATOR

The graph of h consists of line segments through $(0,1)$, $(2,5)$, $(4,4)$ and $(6,0)$. What is the average rate of change of h on $[0,4]$?



(A) $\dfrac{3}{4}$
(B) 1
(C) 2
(D) $-\dfrac{1}{2}$

Solution

Answer: (A). From the graph, $h(0) = 1$ and $h(4) = 4$, so $\dfrac{4-1}{4-0} = \dfrac{3}{4}$.

Question 6 (Free response)

NO CALCULATOR

The position of a cart, in meters, is $s(t) = t^2 - 4t + 10$ for $0 \le t \le 6$ seconds.

(a) Find the average velocity (average rate of change of s) on $[0,4]$. What does your answer mean about the cart?
(b) Find the average velocity on $[4,6]$, with units.
(c) Use the interval $[1,\ 1.01]$ to decide whether the position is increasing or decreasing at $t = 1$.

Solution

(a) $s(4) = 10$ and $s(0) = 10$, so $\dfrac{10-10}{4} = 0$ m/s. The cart is back at its starting position at $t = 4$.
(b) $s(6) = 22$, so $\dfrac{22-10}{2} = 6$ meters per second.
(c) $s(1) = 7$ and $s(1.01) = 6.9801$, so $\dfrac{6.9801-7}{0.01} = -1.99 < 0$: the position is decreasing at

$t = 1$.